

Channel Packer User Guide

For Version 1.0.1

Installation Guide

Itch.io (exe version)

1. Download the tool from the following:
<https://be-a-fish-studios.itch.io/spiders-channel-packing-tool>
2. Press 'Download now' and 'No thanks' to the donation
3. Unzip the folder and open the .exe file.
4. If you get any errors, you need the Channel_Packer_Tool folder in the same directory as the executable file, or it won't run.

Spiders Channel Packing Tool

This tool is a standalone channel packing tool used to merge separate RGB(A) images into a single image and unpack images into separate RGB(A) channels. This is a common texture optimisation technique. It was made to streamline PBR workflows.

This tool is FREE AND OPEN SOURCE! Developed as a uni project for my MSc. Only donate if you want to support me. You can use this for free on commercial projects. You do not need to credit me. (though that would be nice.) You can also modify it if you want. The source code and an alternative download are available on my GitHub.

https://github.com/be-a-fish/Spiders_Channel_Packer

No AI was used in the creation of this software. I do NOT give my consent for it to be used in AI training.

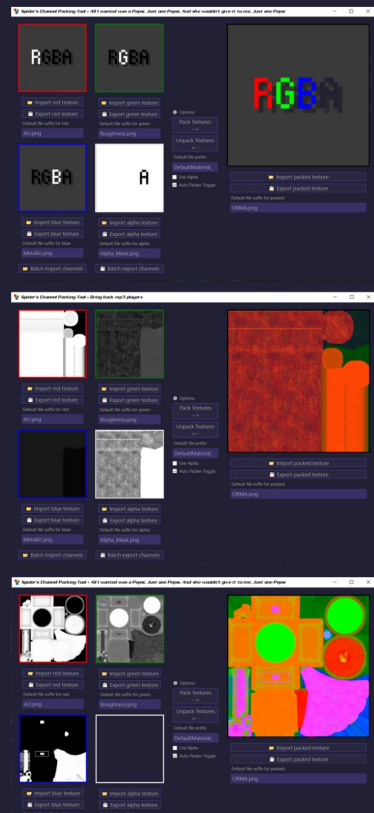
[More information](#) v

Download

Download Now Name your own price

Click download now to get access to the following files:

Spiders Channel Packer 1.0.zip 64 MB



Github (exe version or source code)

1. Go to https://github.com/be-a-fish/Spiders_Channel_Packer and look for the latest release on the right hand side.
2. Download the zip for the exe or source code if you plan on opening it through Python.
3. If opening the Python version, you may need to install some libraries.

 **0** stars

 **0** watching

 **0** forks

Releases 1

 **release 1.0.0** Latest

5 hours ago

Packages

No packages published

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release 1.0.0 Latest

Compare  

 **R-Luff-Escape-Studios** released this 5 hours ago

· 4 commits to main since this release  exe  bed472c

Added a functional executable.

I would like to optimise code a bit further in the future, as 64MB (while being my lucky number) is larger than I want for a lightweight program. I'm sure there's some unnecessary code in there too. That being said, I need to allocate my time accordingly and need to prioritise the user guide, which will be in the next release

Assets 3

 **Spiders.Channel.Packer.1.0.zip** 64.9 MB 6 hours ago

 **Source code** (zip) 6 hours ago

 **Source code** (tar.gz) 6 hours ago



UI Layout

Basic layout

The tool is split into 3 sections: the RGBA channels to the left, the packed image to the right, and the options section in the middle.

The batch options are located under the channels

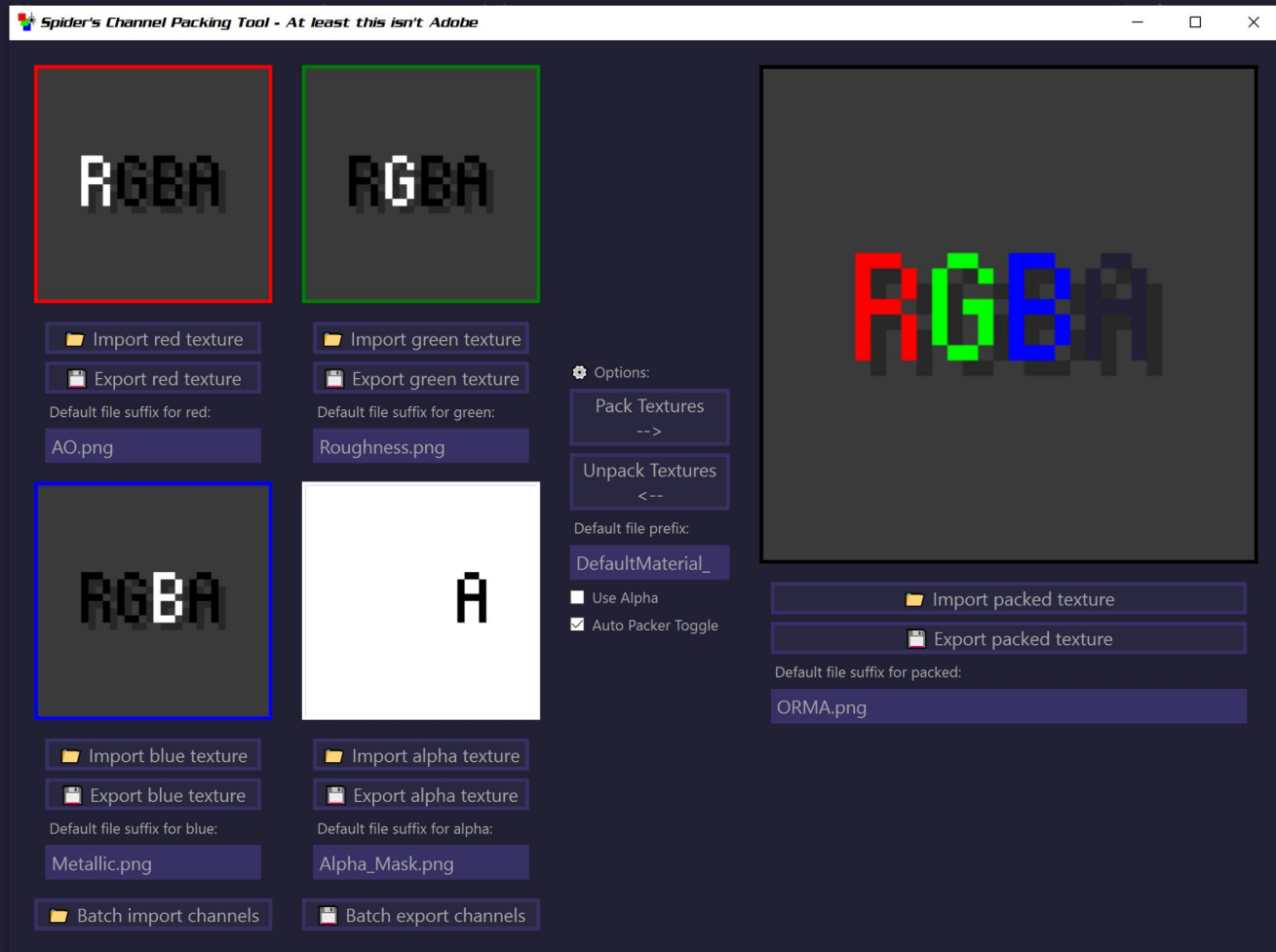


Image channel layout

Each image has the same layout, so the user only has to learn it once. This includes the packed image.

Default file name suffix for selected channel: Added to the prefix for batch importing/exporting



Image texture:
The texture used for the channel.
Outlined in its corresponding colour

 Import red texture

 Export red texture

Default file suffix for red:

AO.png

Import button:
Opens import window

Export button:
Opens export window

Options layout

The options menu contains the basic user options for the tool which apply to all channels.

Default file Prefix: added to the suffix in batch import/export options

Use Alpha toggle: when disabled, alpha channel is not packed and image is exported in RGB colour mode



Options:

Pack Textures

-->

Unpack Textures

<--

Default file prefix:

DefaultMaterial_



Use Alpha



Auto Packer Toggle

Pack texture button:
Packs RGB(A) channels into the packed texture slot

Unpack texture button:
Unpacks the packed texture into the RGB(A) texture slots

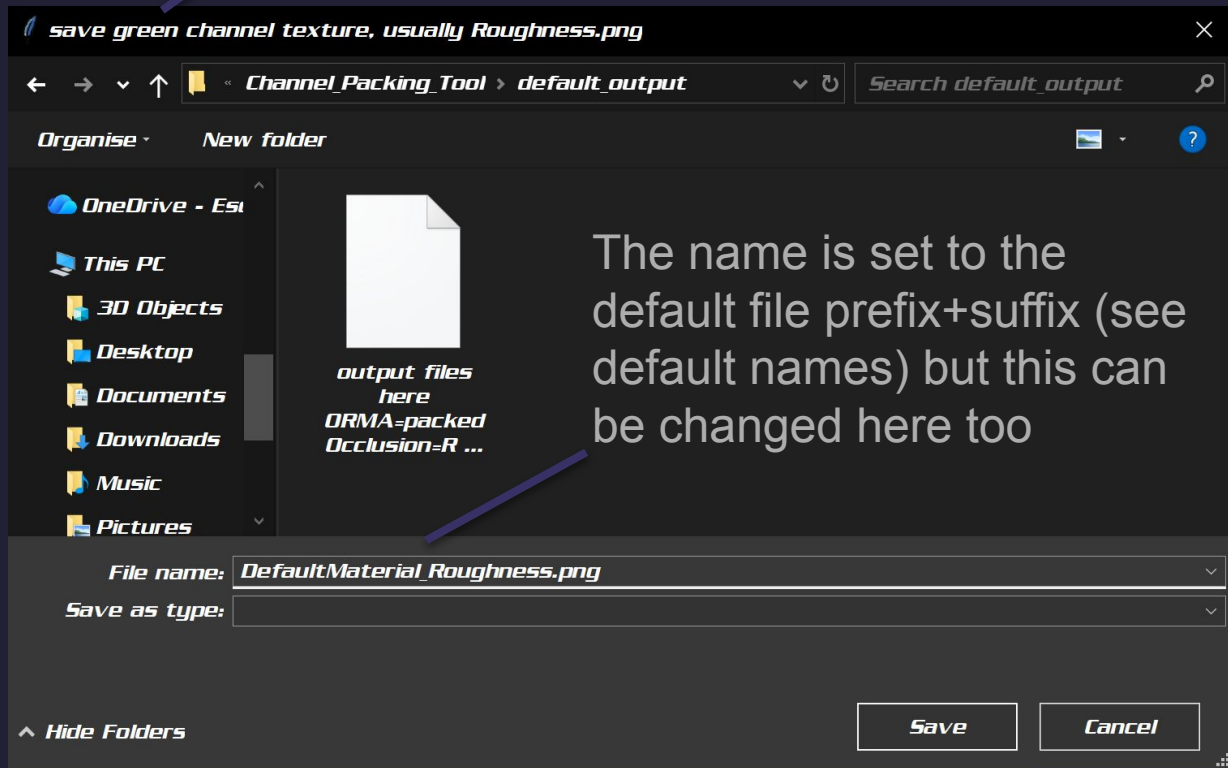
Auto Packer Toggle:
When enabled automatically packs and unpacks textures on import

Export window

The save window should match most other software. The file-type box is currently non-functional, so the user needs to type out their desired extension, but it defaults to .png if left blank.

See also Default path in troubleshooting if you press cancel or X

Window title gives a hint on the channel being saved



Default names – for advanced users

Each filename will be saved with a user-defined name on export; however, advanced users may need to export multiple files with a set format for their pipeline. For example, `T_WeaponName_Roughness.png`.

This is not fully standardised, but most pipelines use an ORM or ORMA (Occlusion, Roughness, Metallic, Alpha) layout, so I've done my best to include relevant default file extensions. A user should choose ones which relate to their project.

The file name is calculated as follows: (Prefix)+(Suffix). Please note the file extension is stored in the suffix string. Using `.jpg` on a 4-channel image will not work, as JPEGs can only store 3 colour channels.

Default file prefix:

DefaultMaterial_

Default file suffix for green:

Roughness.png

Batch Import/Export – for advanced users

Under the separated channels you'll notice batch import and export buttons. These open a folder selection screen. Once you select the folder, they'll import or export based on the file name prefix and suffix set by the user.

For example, if the default prefix is set to “WeaponTexture_” and the green suffix is set to “Roughness.png”, then the batch importer will look for a file named “WeaponTexture_Roughness.png” to put in the green channel.

If the files are missing or named incorrectly, they won't import. If alpha is disabled, it won't look for the file and won't try to import a texture.

As of version 1.0.1, batch import is currently incompatible with the autopacker. This is a bug I plan on fixing soon.



Batch import channels



Batch export channels

Common Use Cases

Packing RGB(A) into a single image

1. Import the unpacked textures using the buttons under the image channels. The open file window name will give you hints of which PBR texture is recommended for the channel.
2. If using the alpha channel, remember to toggle 'Use Alpha'.
3. If autopack is disabled, press Pack Textures.
4. Export the packed texture using the button under the packed image.

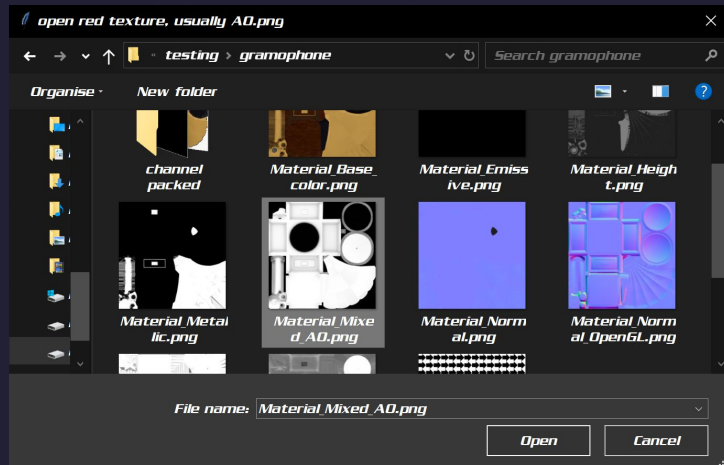


Import red texture

Export red texture

Default file suffix for red:

AO.png



Unpacking RGB(A) from prepacked texture

1. Open a packed image using the 'import packed texture' button under the packed image on the right.
2. If autopacker is disabled, click 'Unpack Textures'.
3. If you want to unpack the alpha, you may need to toggle the 'Use Alpha' tickbox and click 'Unpack Textures'.
4. Export the individual channels using the export button located under the image.
5. Alternatively, set the default file prefix and suffixes and use the batch export button to export them all at once

Adding a single channel to an existing image

This is often used to add an alpha mask to an existing image

1. Open a texture in the packed texture slot and unpack if autopacker is disabled.
2. If you want to add the alpha texture, toggle 'Use Alpha' to true.
3. Replace the individual channel using the import button under the image and repack if necessary.
4. Export the packed texture

Troubleshooting

Tooltips

If you're unsure what a button does, hover your mouse over it for a tooltip.

An example is shown below



Options:

Pack Textures

Takes R G B A channels and merges them into a single texture



Default output

When there are any errors with exporting a texture to a selected location, the tool will automatically save to a default output folder. The file is saved in:

tool_location/Channel_Packing_Tool/default_output and will use the file prefix and suffix assigned by the user. This is a fallback location, so it's worth checking if you get any crashes etc.

Textures not packing

Textures with different resolutions will not pack together. If you're not using the alpha, then consider disabling it and see if it was failing to pack because the unused alpha was the wrong resolution.

Known issues – see bug tracker for more information

GUI: Images scaling breaks aspect ratio

Priority: Low

UPDATE: Kinda works now but jittery and stretches when made smaller.

changed priority to low as it's functional

Aspect ratio forced 1:1 so different images will display as square. chosen to keep it this way for visual consistency still correct size when exporting. only affects GUI

Tool: Images of different resolutions can't pack together

Priority: Medium

Tool: Alpha still unpacking when disabled

Priority: Low

Tool: Gives an error message when attempting to export a .jpg file with 4 Channels

Priority: Medium

GUI: Cursor isn't updated to hand pointer when hovering over a button

Priority: Low

GUI: Save/open windows don't close when closing the main windows

Priority: Low

Tool: Autopacker doesn't work on batch import

Priority: Medium

GUI: Import and export windows don't use the main icon

Priority: Low

Credit

Special Thanks:

Omar Zohdi – for teaching me Python and accommodating my learning differences where applicable.

Escape Studios user testers – for their invaluable feedback and in-person reactions.

Studio 694 (WCUC) – for more invaluable feedback

Leamington Game Dev Group – for even more invaluable feedback

AI Statement

While I actively avoided using any generative AI in my code for moral, ethical and environmental reasons, much of the PySide6 documentation was automatically translated from C, which may have used AI. Additionally, when documentation wasn't enough, I often checked Reddit or other internet forums for help. I can't verify if these sources used AI or not, but I actively avoided ones outright admitting to AI use.

I have done my best to avoid AI in all aspects of this project and can confidently claim this is my own work.

Libraries used:

- Sys
- Os – for paths
- Random – for splash text
- Pyside6 – for GUI
- Functools – to pass variables through button inputs
- Pillow – for image manipulation
- Tkinter – for built-in file open and save windows